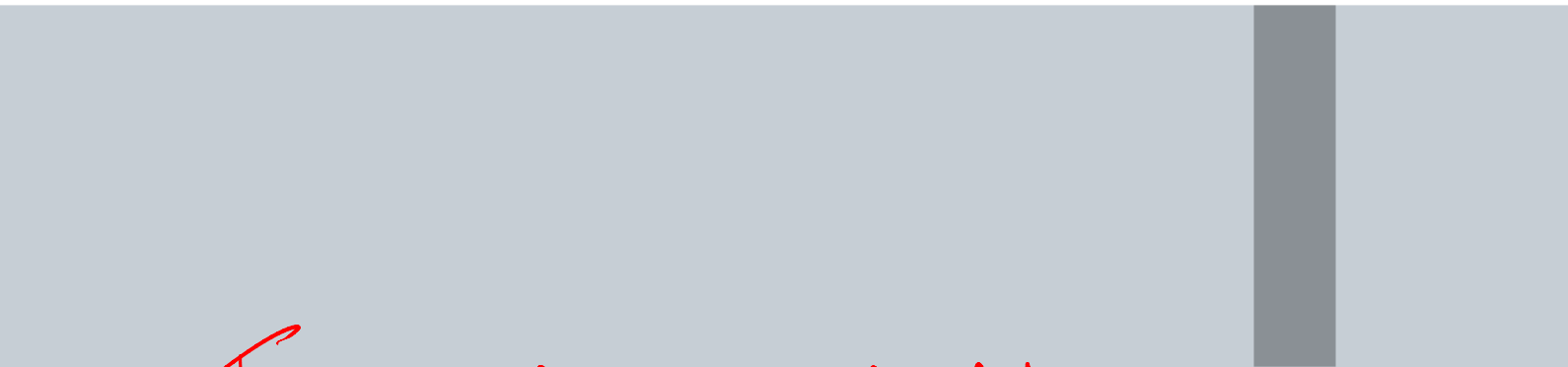
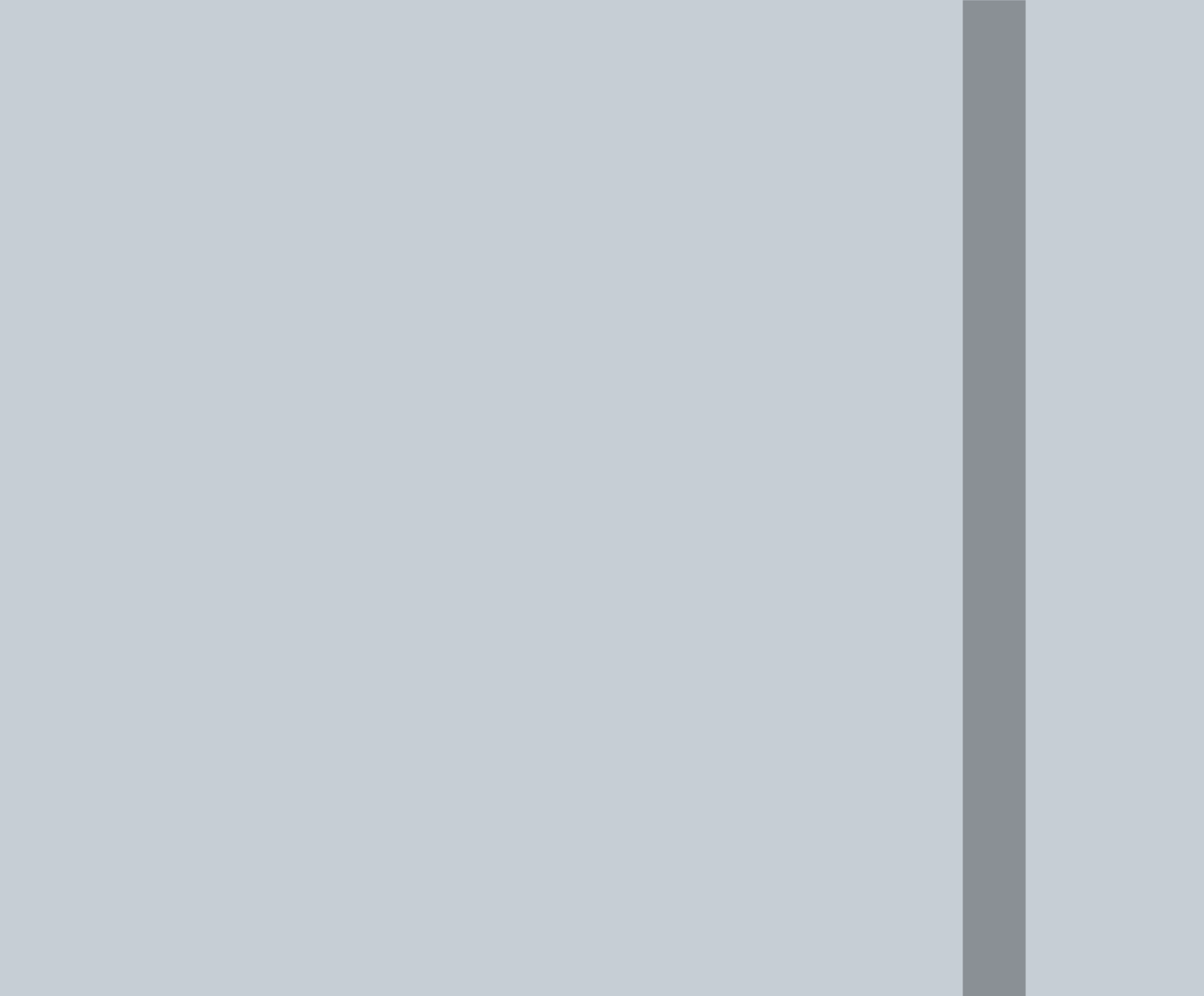




Time, Work and Money

Work done = W , Time taken = T & Efficiency = E

So, $W = E \times T \rightarrow$ like $d = S \times T$ (distance = $S \cdot T$)

$E = \frac{W}{T}$, $T = \frac{W}{E}$ etc. (work done is always 1)

$T = \frac{1}{E}$
Since $W=1$

CONCEPT

$$S = \frac{D}{T} \quad T = \frac{D}{S} \quad D = S \times T$$

$$E = \frac{W}{T} \quad T = \frac{W}{E} \quad W = E \times T$$

$$A \rightarrow 10d \quad B \rightarrow 20d$$

$$E_A = \frac{1}{10} \quad E_B = \frac{1}{20}$$

$$E_{AB} = \frac{1}{10} + \frac{1}{20} = \frac{2+1}{20} = \frac{3}{20}$$

$$T = \frac{1}{\frac{3}{20}} = \frac{20}{3} = 6.66$$

$(W=1=MT)$

$$\frac{M_1 T_1}{W_1} = \frac{M_2 T_2}{W_2}$$

$$\frac{M_1 T_1}{W_1} = 1 = \frac{M_2 T_2}{W_2}$$

$T = \frac{1}{E}$

Q1. A can do a piece of work in 20 days and B can do it in 15 days. How long will they take if both work together?

- A) 62/7 days ☒ B) 60/7 days
C) 66/7 days ☐ D) 67/7 days

$$E_A = \frac{1}{20} \quad E_B = \frac{1}{15}$$

$$\frac{1}{20} + \frac{1}{15} = \frac{3+4}{60} = \frac{7}{60}$$

Q2. Jai can do a piece of work in 10 days and Veeru can do the same work in 20 days. With the help of Basanti, they finish the work in 5 days. How long will it take for Basanti alone to finish the work?

- A) 20 days ☒ B) 10 days
C) 35 days ☐ D) 15 days

$$E_J = \frac{1}{10} \quad E_V = \frac{1}{20}$$

$$\frac{1}{10} + \frac{1}{20} + E_B = \frac{1}{5}$$

$$E_B = \frac{1}{5} - \frac{1}{10} - \frac{1}{20} = \frac{1}{20}$$

Q5. 6 men can do a piece of work in 12 days. How many men are needed to do the work in 18 days?

- A) 3 men ☐ B) 6 men ☐
C) 4 men ☒ D) 2 men ☐

$$\frac{M_1 T_1}{W_1} = \frac{M_2 T_2}{W_2}$$

$$\frac{6 \times 12}{1} = \frac{x \times 18}{1}$$

Q6. X number of men can finish a piece of work in 30 days. If there were 6 men more, the work could be finished in 10 days less. The original number of men is

- A) 10 ☐ B) 11 ☐
C) 12 ☒ D) 15 ☐

$$\frac{x \times 30}{1} = \frac{(x+6) \times 20}{1}$$

$$3x = 2x + 12$$

$$x = 12$$

Q7. Chandler can do a piece of work in 10 days. Joey can do it in 15 days. The total wages for the work is Rs.50. How much should Chandler be paid if they work together for the entire duration of the work?

- A) Rs.30 ☒ B) Rs.20 ☐ C) Rs.50 ☐ D) Rs.40 ☐

$$E_C = \frac{1}{10} \quad E_J = \frac{1}{15}$$

$$\frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$$

$$W_C = \frac{1}{10} \times 6 = \frac{3}{5}$$

$$M_C = \frac{3}{5} \times 50 = 30$$

	C	J
E	$\frac{1}{10}$	$\frac{1}{15}$
	15	10
	3	2
M	$\frac{3}{5} \times 50 = 30$	

Shortcut

Q8. A and B undertake a piece of work for Rs.100. A can do it in 5 days and B can do it in 10 days. With the help of C, they finish it in 2 days. How much should C be paid for his contribution?

- A) Rs.40 ☒ B) Rs.20 ☐ C) Rs.60 ☐ D) Rs.30 ☐

$$E_A = \frac{1}{5} \quad E_B = \frac{1}{10}$$

$$W_A = \frac{1}{5} \times 2 = \frac{2}{5} \quad W_B = \frac{1}{10} \times 2 = \frac{1}{5}$$

$$M_A = \frac{2}{5} \times 100 = 40 \quad M_B = \frac{1}{5} \times 100 = 20$$

$$M_C = 100 - 40 - 20 = 40$$

Q12. If A & B can do a job in 8 days and B & C can do the same job in 12 days. If A, B & C work together they can finish the job in 6 days. How many days can A & C finish the job?

- A) 8 days ☐ B) 10 days ☐ C) 12 days ☐ D) 14 days ☐

$$E_A + E_B = \frac{1}{8} \quad \text{--- (1)} \quad \text{--- (3) - (2)}$$

$$E_B + E_C = \frac{1}{12} \quad \text{--- (2)} \quad E_A = \frac{1}{6} - \frac{1}{12} = \frac{1}{12}$$

$$E_A + E_B + E_C = \frac{1}{6} \quad \text{--- (3)} \quad E_A + E_C = \frac{1}{12} + \frac{1}{24}$$

$$\text{--- (3) - (1)} \quad = \frac{2+1}{24} = \frac{3}{24} = \frac{1}{8}$$

$$E_C = \frac{1}{6} - \frac{1}{8} = \frac{1}{24}$$

Q13. A can do a piece of work in 20 days. He works at it for 5 days and then B finishes it in 10 more days. In how many days will A and B together finish the work?
 A) 8 days B) 10 days C) 12 days D) 6 days

→ Sol $\Rightarrow E_A = \frac{1}{20}$, $W_A = EXT = \frac{1}{20} \times 5 = \frac{1}{4}$

So, $W_{\text{remaining}} = \frac{3}{4}$

So, $E_B = \frac{W_B}{T_B} \Rightarrow E_B = \frac{3/4}{10} \Rightarrow E_B = \frac{3}{40}$

So, $E_A + E_B = \frac{1}{20} + \frac{3}{40} = \frac{2+3}{40} = \frac{5}{40} = E_{AB}$

Hence $T_{AB} = 1/E_{AB} = \frac{40}{5} = \underline{\underline{8 \text{ days}}}$

Q15. Twenty workers can finish a piece of work in 30 days. After how many days should 5 workers leave the job so that the work is completed in 35 days?
 A) 5 days B) 10 days C) 15 days D) 20 days

→ $20 \times 30 = 5 \times x + 15 \times 35$ ($\frac{M_1 T_1}{W_1} = \frac{M_2 T_2}{W_2}$)
 $\Rightarrow x = 15$

Q9. 4 men and 3 women finish a job in 6 days, and 5 men and 7 women can do the same job in 4 days. How long will one man and one woman take to do the work?
 A) 156/7 days B) 51/2 days C) 58/7 days D) 271/22 days

$\frac{M_1 T_1}{W_1} = \frac{M_2 T_2}{W_2} = \frac{M_3 T_3}{W_3}$
 $\frac{(4M+3W)6}{W_1} = \frac{(5M+7W)4}{W_2} = \frac{(M+W)x}{W_3}$
 $(4M+3W)6 = (5M+7W)4 = (M+W)x$
 $12M+18W = 20M+28W = (M+W)x$
 $12M+18W = 20M+28W$
 $8M = 10W$
 $M = \frac{5}{2}W$
 $(5 \times \frac{5}{2}W + 7W)4 = (M+W)x$
 $(\frac{25}{2}W + 7W)4 = (\frac{5}{2}W + W)x$
 $(\frac{39}{2}W)4 = (\frac{7}{2}W)x$
 $39 \times 4 = 7x \Rightarrow x = \frac{156}{7}$
 (Imp. que)
 (lengthy so do it in last if time left)

Q10. 5 men and 3 boys can together cultivate a 23-acre field in 4 days and 3 men and 2 boys together can cultivate a 7-acre field in 2 days. How many boys will be needed together with 7 men, if they have to cultivate a 45-acre field in 6 days?
 A) 2 B) 3 C) 4 D) 6

$\frac{(5M+3B)4}{23} = \frac{(3M+2B)2}{7} = \frac{(7M+xB)6}{45}$
 $(5M+3B)4 \times 7 = (3M+2B)2 \times 23$
 $70M + 42B = 69M + 46B$
 $M = 4B$
 $(3 \times 4B + 2B)15 = (7 \times 4B + xB)7$
 $14B \times 15 = (28+x)B \times 7$
 $80 = 28+x$
 $x = 2$
 → same concept as left que

Q16. Gabbar and Mogambo together can do a piece of work in 7 days. If Gabbar does twice as much work as Mogambo in a given time, how long will Gabbar alone take to do the work?
 A) 6.33 days B) 10.5 days C) 11 days D) 72 days

$E_G + E_M = \frac{1}{7}$
 $E_G = 2E_M$
 $E_M = \frac{E_G}{2}$
 $E_G + \frac{E_G}{2} = \frac{1}{7}$
 $\frac{3}{2}E_G = \frac{1}{7}$
 $E_G = \frac{1}{7} \times \frac{2}{3} = \frac{2}{21}$
 $T = \frac{21}{2} = \underline{\underline{10.5}}$

Q17. The ratio of efficiencies of P, Q and R is 2 : 3 : 4. While P and R work on alternate days, Q works on all days. The work is completed in 10 days and the total amount they get is Rs. 1200. Find the amount earned by each person (respectively).
 A) 200, 600, 400 B) 400, 600, 200 C) 600, 200, 400 D) 400, 200, 600

	P	Q	R
Eff	2	3	4
Time	5	10	5
	1 : 2 : 1		
W	2x1	3x2	4x1
	1 : 3 : 2		

Q18. Bruce can copy 50 pages in 10 hours. Bruce and Robin together can copy 300 pages in 40 hours. In how much time can Robin copy 30 pages?

A) 13 hours ☒ B) 12 hours C) 11 hours D) 9 hours

$$E_B = \frac{50}{10} = 5$$

$$E_B + E_R = \frac{300}{40} = \frac{15}{2}$$

$$E_R = \frac{15}{2} - 5 = \frac{5}{2}$$

$$T_R = \frac{30}{\frac{5}{2}} = 12$$

B	50p	10 hrs
B+R	300p	40 hrs
B	200p	40 hrs
R	100p	40 hrs
	10p	4 hrs
	30p	12 hrs

Q19. There are 720 boxes. A & B can paint them in 20 days, B & C can paint them in 24 days, A & C in 15 days. If A paints for 4 days, B for 8 days and C for 8 days, how many boxes will be painted?

A) 252 B) 516 ☒ C) 348 D) 492

$$E_A + E_B = \frac{720}{20} = 36 \quad \text{--- (1)}$$

$$E_B + E_C = \frac{720}{24} = 30 \quad \text{--- (2)}$$

$$E_C + E_A = \frac{720}{15} = 48 \quad \text{--- (3)}$$

$$W_B + W_C = 30 \times 8 = 240$$

$$\text{(1) + (3) - (2)}$$

$$2E_A = 36 + 48 - 30 = 54$$

$$E_A = \frac{54}{2} = 27$$

$$W_A = 27 \times 4 = 108$$

$$\text{Total} = 240 + 108 = 348$$

Q20. Sansa can do a piece of work in 10 days, Arya in 12 days and Bran in 15 days. They all start the work together, but Sansa leaves after 2 days and Arya leaves 3 days before the work is completed. In how many days is the work completed?

A) 2 days B) 6 days ☒ C) 7 days D) 8 days

$$E_S = \frac{1}{10} \quad E_A = \frac{1}{12} \quad E_B = \frac{1}{15}$$

$$W_S + W_A + W_B = 1$$

$$\frac{1}{10} \times 2 + \frac{1}{12} \times (x-3) + \frac{1}{15} \times x = 1$$

$$\frac{12 + 5x - 15 + 4x}{60} = 1$$

$$9x - 3 = 60$$

$$9x = 63$$

$$x = 7$$



